TRIAGE-LINK — Productization Gap Backlog (Prototype → Commercial Product)

> **Purpose.** The concrete engineering work to turn the prototype into a sellable product, **sized** and **sequenced**, with the **commercial tier** each gap unlocks. Pairs with `docs/commercialization-strategy.md` (which tier/segment needs what) and `docs/CERTIFICATION-ROADMAP.md` (the regulated-market evidence). One of three exploration deliverables — read alongside the strategy and the regulatory brief before deciding.
>
> *Sizing:* **S** ≈ days · **M** ≈ 1–2 wks · **L** ≈ 3–6 wks · **XL** ≈ quarter+. *Phase* maps to the strategy's commercial roadmap (P1 beachhead → P2 scale niche → P3 regulated/EMS).
>
> **North star constraint:** offline-first and **no-backend stays the default**. Every paid/online feature must **degrade gracefully** to the offline core — that's the moat, not a checkbox.

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Where the prototype stands today

Built: offline PWA, IndexedDB store + op-log, body chart, vitals/trends, START triage board, scene summary, AT-MIST card, FHIR R4 export, EN/FR/AR/FA (RTL), opt-in at-rest vault + encrypted backups, CSP/API hardening, CI scanning. **Single-device, single-user, no accounts, no server required.**

The gaps below are what stands between that and "a customer's IT/clinical/procurement team says yes."

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1. Identity & access

Gap	Why it's needed	Size	Phase	Unlocks
User accounts + sign-in	Attribute records to a clinician; baseline for audit	M	P1	Team
Multi-tenant org model	Separate data per NGO/agency/event; admin boundaries	L	P1	Team/Org
Role-based access control (RBAC)	Field / lead / admin scopes; least privilege	M	P1	Org
SSO (OIDC/SAML)	Enterprise/agency procurement requirement	L	P2	Org/Enterprise
Device provisioning / enrolment	Hand out configured installs to a field team	M	P2	Org

2. Security & compliance hardening

Gap	Why	Size	Phase	Unlocks
Always-on encryption policy (org-enforced, not opt-in)	Real-PHI handling; remove the "off by default" risk	M	P1	Org
Key management & recovery (org escrow / rotation)	Avoid lost-passphrase data loss at org scale	L	P1	Org
Immutable audit log (who-saw/changed-what)	PHIPA/HIPAA-grade accountability; procurement gate	L	P1	Org/EMS
Retention & destruction policy + tooling	Health-privacy compliance	M	P2	Org
SOC 2 Type II readiness	Enterprise/government trust	XL	P2	Enterprise
Independent pen test + vuln-disclosure program	Sector expectation; de-risks deals	M	P2	Org/Enterprise

3. Data, sync & reliability

Gap	Why	Size	Phase	Unlocks
Productionize `sync-service` (auth, multi-tenant, rate limits)	Optional shared/team data without losing offline-first	XL	P2	Org (optional)
Conflict-resolution hardening at scale	Op-log resolver proven beyond single device	L	P2	Org
Backup / disaster recovery (hosted tier)	Uptime & data-durability guarantees	L	P2	Org/Enterprise
Observability (logging, metrics, error reporting)	Operate a service; support SLAs	M	P2	Org
Uptime SLA + status page	Contractual requirement for hosted tiers	S	P2	Org/Enterprise

4. Interoperability & data standards

Gap	Why	Size	Phase	Unlocks
FHIR conformance hardening (CA-Core / profiles, validation)	Credible standards-based handover	L	P2	Org/EMS
OADS v4.0 dataset conformance	*Mandatory* for official Ontario ACR use	XL	P3	EMS
NEMSIS dataset/export	US + national EMS data flows	XL	P3	EMS (US)
CAD / dispatch integration	Incumbent-parity feature for EMS	XL	P3	EMS
Hospital EHR exchange (Epic/Oracle/etc.)	Closes the handover loop for EMS	XL	P3	EMS

5. Admin, ops & monetization

Gap	Why	Size	Phase	Unlocks
Org admin console (users, roles, settings)	Self-serve management	L	P1	Org
Reporting / analytics (roster, exports, basics)	Coordinator/QA value	M	P2	Org
Data import / export / migration	Avoid lock-in fear; onboard from paper/other tools	M	P1	Team/Org
Licensing / entitlement enforcement	Gate paid tiers (open-core boundary)	M	P1	Team/Org
Billing / subscription management	Collect revenue	M	P2	Team/Org

6. UX, accessibility & distribution

Gap	Why	Size	Phase	Unlocks
Accessibility (WCAG) pass	Procurement + field usability (gloves/sunlight)	M	P1	All
Language extensibility (add a language w/o code release)	Direct selling point for multilingual buyers	M	P1	All
White-label / branded builds	NGO/agency branding	S	P1	Team/Org
MDM / managed-device guidance + packaging	Fleet deployment	M	P2	Org
Install/distribution polish (stores, kiosk, signed builds)	Trust + ease of rollout	M	P1	All

7. Support & documentation (product, not code)

Gap	Why	Size	Phase	Unlocks
Onboarding + training material	Adoption in non-technical field teams	M	P1	All
Support process + SLA tiers	Required for paid tiers	M	P1	Team/Org
Admin/clinical docs + runbooks	Reduce support load; procurement evidence	M	P1	All

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Phase rollup (what "done" means)

- **P1 — beachhead-ready:** accounts, multi-tenant, RBAC, always-on crypto + key recovery, immutable audit, import/export, entitlements, accessibility, language extensibility, onboarding/support. *This is the minimum to deploy real PHI for a paying NGO/event customer.*
- **P2 — scale the niche:** SSO, hosted sync + DR + observability + SLA, reporting, billing, SOC 2, pen test, FHIR conformance.
- **P3 — regulated/EMS:** OADS/NEMSIS conformance, CAD/EHR integration, plus the SaMD evidence stack in the cert roadmap.

Biggest, riskiest items (watch these)

1. **Multi-tenant + hosted sync** (XL) — the architectural shift from single-device; do it without compromising offline-first.
2. **OADS/NEMSIS conformance** (XL each) — the procurement moat; only worth it for the EMS channel (P3).
3. **SOC 2** (XL, mostly process) — long lead time; start early if enterprise/gov is targeted.

> **Sequencing principle:** P1 is small and segment-driven — you do *not* need the XL items to earn first revenue in the humanitarian/event niche. Defer every P3 item until the EMS channel is a deliberate, funded decision.